Tuan Nguyen

Ph.D. Candidate | Interactive ML | Math & Logic

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Education

University of Arizona, Tucson, AZ Aug 2022 – Present
Ph.D. in Computer Science (Expected Spring 2027)
Advisor: Dr. Kwang-Sung Jun and Dr. Chicheng Zhang
Research: Interactive ML, Reinforcement Learning, Monte Carlo Tree Search, Large Language Model Reasoning

University of Oregon, Eugene, OR
M.S. in Computer Science
Advisor: Dr. Thien Huu Nguyen
Research: Natural Language Processing, Information Extraction, Domain Adaptation, Multilingual NLP

CUNY Baruch College, New York, NY
B.A. in Finance; minor in Mathematics

Research Interests

- My research is on interactive machine learning, where learning agents actively engage in data collection and make decisions to pursue objectives within an environment. We develop methods that guide exploration toward the most relevant parts of the environment, thereby reducing data collection costs and improving learning efficiency.
- I am currently focused on developing practical machine learning methods that enhance the reasoning capabilities of large language models (LLMs).

Research Assistant

Technical Experience

- **LLM reasoning models:** Implemented tree-search algorithms, including Beam Search and Monte Carlo Tree Search, for large language model reasoning. Worked with multiple LLM families and reward models, including LLaMA, Qwen, Phi, and Mistral, and evaluated them on mathematical reasoning benchmarks such as GSM8K, MATH, and AIME.
- **LLM experimentation:** Ran large language model training and inference experiments on multi-GPU clusters, and tracked and managed experiments using weights & biases. Applied tree-search algorithms to generate and select high-quality reasoning steps for downstream fine-tuning.
- **Parallel experimentation framework:** Developed a multi-threaded framework to support repeated trials of reinforcement learning experiments, improving overall efficiency and scalability.

Diverse Exploration for Reasoning in Large Language Models (In Progress)

- Designed DiverseExp, a Monte Carlo Tree Search (MCTS) operator that enhances exploration to find diverse, high-quality reasoning trajectories, improving both test-time inference and synthetic data generation for downstream finetuning.
- Consistently outperformed prior baselines when evaluated on math reasoning benchmarks (e.g., GSM8K, MATH, AIME) across multiple LLM families (e.g., LLaMA, Qwen, Phi, Mistral).

Error Reduction for Q-Value Estimation in Reinforcement Learning (Published in AISTATS 2025)

- Developed HAVER, a novel estimator that minimizes mean squared error (MSE) in Q-value estimation, effectively improving the learning process in reinforcement learning algorithms such as Q-learning and MCTS.
- Established the first instance-dependent MSE guarantees, proving that HAVER not only matches the traditional max estimator in the worst case but substantially outperforms it across many problem instances.
- Consistently outperformed established estimators (max, double, and weighted) across controlled multi-armed bandit and toy environments (Gridworld, Frozen Lake), significantly reducing estimation error and improving accumulated rewards.

Publications (Reverse Chronological)

*Co-first authors; authors contributed equally.

HAVER: Instance-Dependent Error Bounds for Maximum Mean Estimation and Applications to Q-Learning and Monte Carlo Tree Search
Tuan Ngo Nguyen, Jay Barrett, and Kwang-Sung Jun
In Proceedings of the AISTATS 2025

Fixing the Loose Brake: Exponential Tail Bounds for Stopping Time in Best Arm Identification
Kapilan Balagopalan*, Tuan Ngo Nguyen*, Yao Zhao*, and Kwang-Sung Jun
In Proceedings of the ICML 2025

Crosslingual Transfer Learning for Relation and Event Extraction via Word Category and Class Alignments
Minh Van Nguyen, Tuan Ngo Nguyen, Bonan Min, and Thien Huu Nguyen
In Proceedings of the EMNLP 2021

Crosslingual Transfer Learning for Relation and Event Extraction via Word Category and Class Alignments
Minh Van Nguyen, Tuan Ngo Nguyen, Bonan Min, and Thien Huu Nguyen
In Proceedings of the EMNLP 2021

Event Detection: Gate Diversity and Syntactic Importance Scores for Graph Convolution Neural Networks
Viet Dac Lai, Tuan Ngo Nguyen, and Thien Huu Nguyen
In Proceedings of the EMNLP 2020

Graph Transformer Networks with Syntactic and Semantic Structures for Event Argument Extraction
Amir Pouran Ben Veyseh, Tuan Ngo Nguyen, and Thien Huu Nguyen
In Findings of the EMNLP 2020

Programming Skills

Programming Languages: Python, C/C++.

ML & LLM Frameworks: PyTorch, Transformers, vLLM, DeepSpeed, LangGraph, SGLang.

Developer Tools & OS: Linux, Git, Emacs, VS Code, Claude.

Academic Services

Conference Reviewer: ACL, EMNLP, IJCAI, COLT.

Teaching Assistant: Design and Analysis of Algorithms, Principles of Data Science.